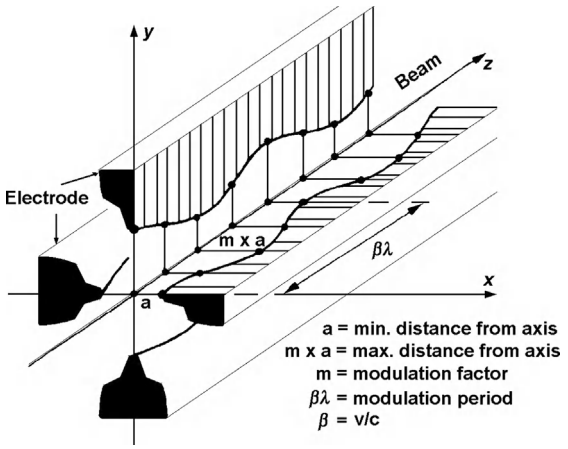


Figure 16.6 RFQ accelerator showing the four shaped electrodes (vanes) of the RF cavity.

electron mass is ~97 800 times greater than its rest mass (a little heavier than a chromium atom).

An important advance in linear accelerator technology is the development of radio-frequency quadrupoles (RFQ:s), permitting either simultaneous acceleration and focusing of high currents of particles or the funnelling of two or more beams into one which is then fed into a linear accelerator operating at a higher frequency. In principle the RFQ is somewhat similar to the Alvarez design but without separate drift tubes, see Figure 16.6. Its main characteristics can be summarized as: i) it is linear accelerator operating in the energy range 20 keV/u to 2 MeV/u, ii) it bunches, focuses, and accelerates particles with RF fields, iii) typical lengths are 1 to 2 m and operating frequencies 50 to 400 MHz, iv) intervane voltage ~100 kV, v) minimum value of *a* (see Figure 16.6) 3–4 mm, Q-factor ~10,000, vi) intervane capacity ~10^{10} F/m. In order to accept an incoming unbunched beam the a-value has to be rather high at the entrance and then progressively decrease to its final constant value. The undulating vane profiles are normally obtained by using numerically controlled milling machines.

16.6. CYCLOTRONS

The difficulties inherent in the length of high energy linear accelerators were ingeniously overcome by Lawrence and Livingston in their construction of the first cyclotron in 1931. The basic operating principles of the cyclotron are shown in Figure 16.7. The particles are accelerated in spiral paths inside two semicircular flat metallic cylinders called dees (D's), placed in a flat vacuum chamber. The two dees are connected to a high frequency alternating voltage. The volume within the dees corresponds to an equipotential condition just as does the volume within the drift tubes of linear accelerators. The

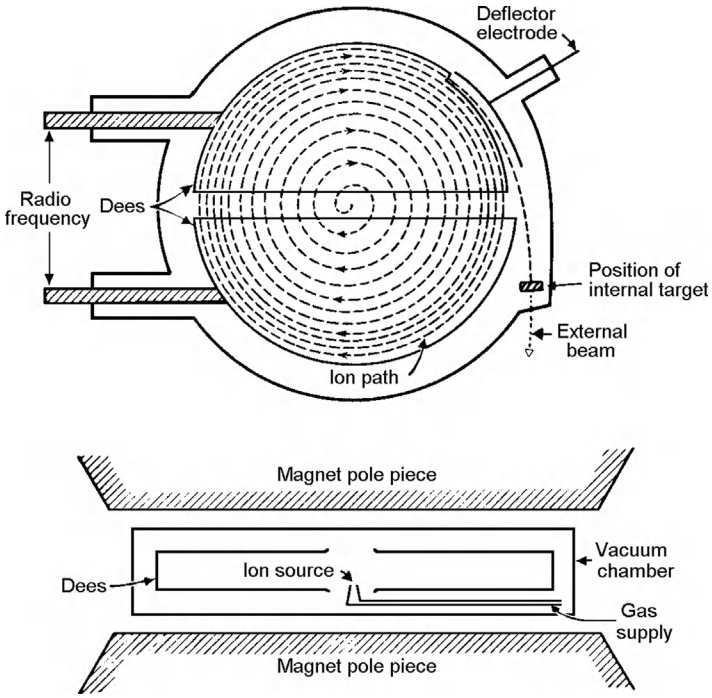


Figure 16.7 Principle of a cyclotron. *(From R.S. Shankland.)*

dees and their vacuum chamber are placed between the two poles of a magnet so that the magnetic field operates upon the ion beam to constrain it to flat circular paths inside the dees. At the gap between the dees the ions experience acceleration due to the potential difference between the dees. This causes an increase in the radius when they enter the next dee. The beam particles originate at the ion source at the center of the cyclotron, and as they spiral outward in the dees they acquire a constant increase in energy for each passage across the gap between the dees. The target is located either inside the vacuum chamber (internal beam) or the beam can be taken out through a port into an evacuated tube or through a thin "window" into the ambient air (external beam) after extraction from the circular path by a deflector.

As the projectiles acquire energy while passing the gaps between the dees, the radius of their path within the dees increases step by step. The longer path-length in the dees is related to the energy of the projectile by the equation (§3.12)

$$E_{\text{kin}} = q^2 r^2 B^2/(2m) \tag{16.8}$$

where r is the radius of the beam path at the energy E_{kin} and B is the value of the magnetic field. This equation is valid for particles of non-relativistic velocities. The successful operation of the cyclotron is based on the principle that the increase in the

radius of the path in the dee nearly compensates for the increase in particle velocity. As a result, the particles, as they spiral through the dees, arrive at the gap when the potential difference is of the right polarity to cause acceleration. The principle of phase stability applies also to the cyclotron.

The frequency (in Hz) requirement is

$$\nu = qB/(2\pi m) \tag{16.9}$$

or

$$\nu = v/(2\pi r) \tag{16.10}$$

where v is the particle velocity at radius r. From (16.9) it follows at constant ν that B/m must also be constant.

The maximum energies available in conventional cyclotrons of the constant frequency type are about 25 MeV for protons and deuterons, and 50 MeV for α-particles. The shape of the electrostatic field at the gap between the dees as well as the design of the magnetic field to produce a slight non-uniformity at the outer edges of the dees produce a focusing effect on the beam particles. The ion currents are usually $\leq$0.5 mA for the internal beam, and about a factor 10 less for an external beam.

As the energies increase, so does the relativistic mass. From (16.8) we can see that either the frequency or the magnetic field, or both, must be modified to compensate for the increasing mass in order to have the beam particles arrive at the gap in phase with the voltage. The former is done in synchrocyclotrons (next section) while the magnetic field is varied in sector focused cyclotrons. In the latter, the magnet has ridges (often of spiral form) which provide sufficient modification of the field (retaining azimuthal focussing despite increasing total field strength) to allow the acceleration of particles to much higher energies than is possible with a conventional cyclotron (Fig. 16.8). The energy limit is at about twice the rest mass of the ions. Superconducting magnets are also used to achieve the highest possible magnetic field.

In the *separated-sector cyclotron*, SSC, the magnet is split into two, four, or more, submagnets, each magnet bending the beam only part of the circle. Between the bending magnets the beam follows a linear path. A few (or none) acceleration gaps (corresponding to the dees) are located in each of the linear paths, see Figure 16.8(a). The split of the magnet into several separate sectors permits a larger bending radius to be used, i.e. a higher energy can be achieved with smaller magnet pole pieces although the total magnet weight normally increases compared to a single magnet machine. As an example, in the GANIL accelerator complex (at Caen, France) normal cyclotrons are used as injectors to a SSC with two acceleration cavities, each corresponding to two acceleration steps, and four bending magnets. The beam exiting from the SSC is further stripped by passage through a thin carbon foil and then fed into a second SSC where it achieves its

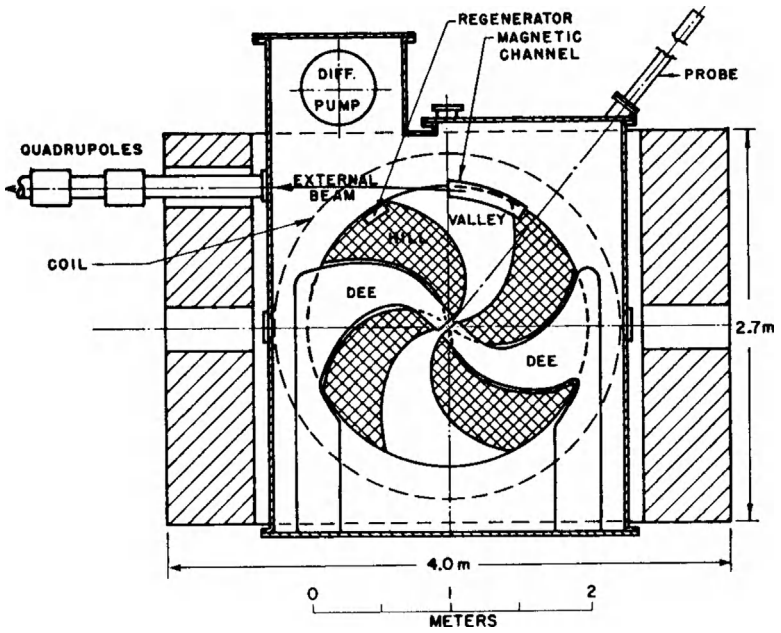


Figure 16.8 A 150 MeV p^+ sector-focused cyclotron showing the magnet with its spiral-ridged pole pieces and ¼-wave D-design.

final energy, see Figure 16.9(b). The beam quality from SSC no 2 is increased by a magnetic spectrometer.

Although there is a limit to the energies achievable with cyclotrons of constant frequency and magnetic field, they have the great advantage of high beam currents. For a heavy ion cyclotron, accelerating particles of charge z ($z \leq Z$), the final energy per mass unit, E/A (MeV/u), is given by

$$E/A = K(z/A)^2 \tag{16.11}$$

and such machines are usually characterized by their K-value. The GANIL SSC:s have e.g. both $K = 400$ but use different z of the ions. The development of ion sources producing higher currents of more highly charged ions is important as it increases the maximum energy achievable at a given K-value.

The magnetic bending of the beam corresponds to a centrifugal acceleration of the charged particles. The acceleration or deceleration of electric charges is always accompanied by the emission of electromagnetic radiation. Although only a small part of the energy input of a cyclotron is lost through this *synchrotron radiation*, it is still sufficient to require considerable radiation shielding (usually concrete walls). Synchrotron radiation is much less of a problem for linear accelerators, except for protons.

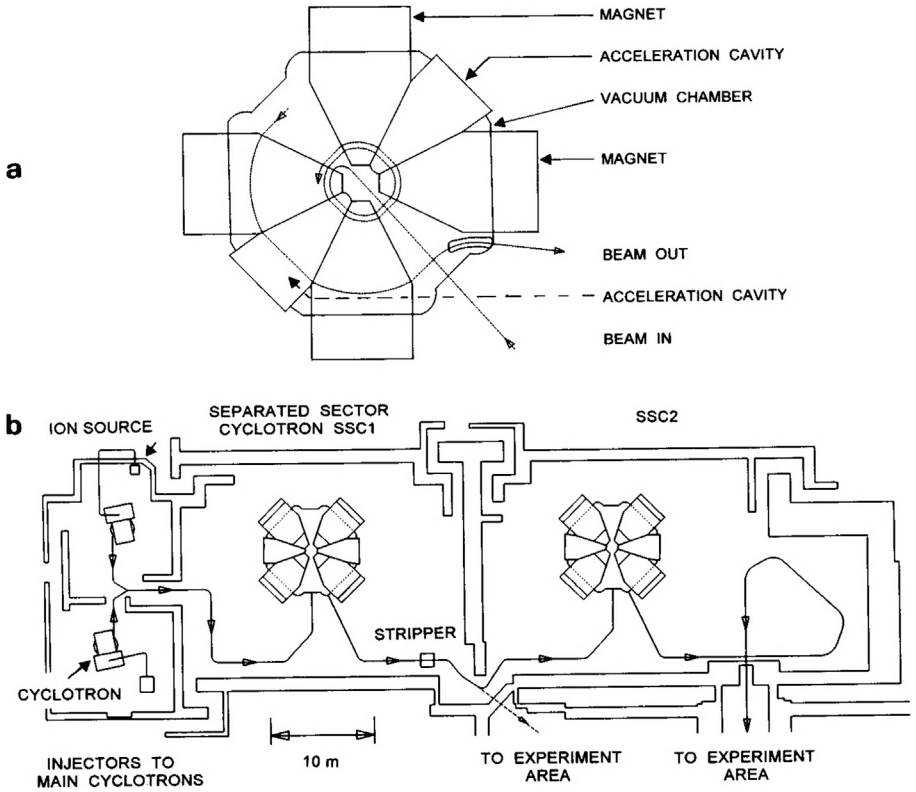


Figure 16.9 (a) Principle of a sector-separated cyclotron and (b) the GANIL accelerator complex (beam-line bending magnets are not shown).

16.7. FREQUENCY MODULATED CYCLOTRONS AND SYNCHROTRONS

The frequency of a cyclotron can be modulated to take into account the variation in velocity and mass as relativistic effects increase in importance. At very high energies the radius becomes very large. This has led to two somewhat different accelerator designs. The *frequency modulated* (FM) or *synchrocyclotron* maintains the original cyclotron principle with a spiral particle path, while in *synchrotrons* the particle path is fixed in a circular orbit; Figure 16.10 is an example of the latter.

In a synchrotron there is a balance between frequency and the magnetic field, one or both being varied during the acceleration (cf. (16.7a)). From (16.6) and (16.7) one obtains

$$E_{\text{kin}} = \nu\pi r^2 Bq \tag{16.12}$$

by eliminating the mass from the relation for the kinetic energy of the projectile. In these machines the particles are accelerated in bursts, since the frequency of the accelerating